

SHOVON MANDAL

Lecturer, Department of Computer Science and Engineering | Northern University of Business and Technology Khulna, Bangladesh
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RESEARCH PROFILE

Computer Science and Engineering lecturer and graduate researcher working on secure distributed systems and applied intelligent computing. My research includes blockchain-based digital payments, NFC and IoT transaction security, IOTA-based decentralized communication, embedded systems security, applied machine learning, natural language processing, and explainable AI. I have authored peer-reviewed conference publications in distributed systems, cybersecurity, machine learning, NLP, and AI-enabled applications, with additional manuscripts currently under review. I am interested in pursuing funded Ph.D. or thesis-based M.S. opportunities aligned with these research areas.

RESEARCH INTERESTS

Secure Distributed Systems; Blockchain-Based Digital Payments; IoT and NFC Transaction Security; Applied Cryptography; Embedded Systems Security; Applied Machine Learning; Natural Language Processing; Explainable AI.

EDUCATION

M.Sc. Eng. in Computer Science and Engineering

Jul 2022 - Present

Khulna University of Engineering and Technology, Bangladesh

Status: Only thesis defense remaining | **Thesis:** Blockchain Technology Based Smart E-Payment Strategy for Secure Cashless Payment

Supervisor: [Dr. Pintu Chandra Shill](#) | **Research Area:** Secure Transaction Verification in Distributed Networks (Blockchain, NFC)

Relevant Coursework: Principles of Cryptography; Advanced Network Security; Cloud Computing; Neural Networks.

B.Sc. in Computer Science and Engineering

Spring 2018 - Spring 2022

BRAC University, Bangladesh

CGPA: 3.80/4.00 | **Academic Standing:** Highest Distinction

Thesis: Blockchain-based Micropayment System for Secured Cashless Small Payments

Supervisor: [Md. Golam Rabiul Alam, PhD](#) | **Research Area:** Applied Cryptography for Secure Distributed Systems

PUBLICATIONS AND RESEARCH

Published Conference Papers

1. **Mandal, S., & Shill, P. C.** (2024). "Ensuring Security and Efficiency in Digital Payments using Blockchain with NFC." *2024 International Conference on Knowledge Engineering and Communication Systems (ICKECS)* [DOI](#).
2. Ghosh, A., **Mandal, S.**, Islam, M. J., Islam, K. M., Dhar, S., & Baidya, R. (2026). "Comparative Evaluation of Multiple CNN Architectures for Dermoscopic Skin Lesion Classification using ISIC Dataset." *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)* [DOI](#).
3. Keya, M., **Mandal, S.**, et al. (2024). "Q-learning based Automated Message Multicast in Gossip Protocol for Node Confirmation in IOTA Tangle." *2024 27th International Conference on Computer and Information Technology (ICCIT)* [DOI](#).
4. Ghosh, A., Pinki, F. T., Baidya, R., Alam, F., & **Mandal, S.** (2026). "A Machine Learning and DEM-Based Approach for Predicting Sea Level Rise and Assessing Coastal Risk in Bangladesh Using AI Insights." *2026 IEEE 15th International Conference on Communication Systems and Network Technologies (CSNT)* [DOI](#).

Accepted / In Press Conference Papers

1. **Mandal, S.**, et al. (2025). "Revolutionizing Academic Learning and Research with the Integration of Leap Motion and ARCore in Android." *16th International Conference on Computing, Communication and Networking Technologies (ICCCNT), IIT Indore, India*.
2. Ghosh, A., **Mandal, S.**, et al. (2025). "Advancing Sentiment Analysis Through Natural Language Processing and Deep Learning." *16th International Conference on Computing, Communication and Networking Technologies (ICCCNT), IIT Indore, India*.

Under Review

1. **Mandal, S.**, Ahmed, K. A., Ghosh, A., Swadesh, P. M., & Baidya, R. (2026). "Data-Driven Multiclass Prediction of Student Academic Outcomes Using Machine Learning: A Comparative Evaluation of SVM and CatBoost." Manuscript submitted to the *5th IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2026)*, Jashore, Bangladesh.
2. Islam, M. H., Jiku, N. A. S., **Mandal, S.**, Swadesh, P. M., & Ghosh, A. (2026). "Automated Visual Inspection of Tilapia Fish Quality Using CNN and Transfer Learning." Manuscript submitted to the *5th IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2026)*, Jashore, Bangladesh.
3. Ghosh, A., Datta, A., **Mandal, S.**, Swadesh, P. M., Ahmed, K. A., & Molla, M. H. (2026). "Beyond Single Transformers: A BanglaBERT-XLM-RoBERTa Stacking Framework for Code-Mixed Emotion Recognition." Manuscript submitted to the *3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)*, Dhaka, Bangladesh.
4. Ghosh, A., Molla, M. H., Swadesh, P. M., **Mandal, S.**, Ahmed, K. A., & Datta, A. (2026). "Chameleon Words in Cross-Domain Sentiment Analysis: An Explainable Framework with LLM-Based Style Transfer." Manuscript submitted to the *3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026)*, Dhaka, Bangladesh.

SELECTED RESEARCH AND DEVELOPMENT PROJECTS

Blockchain and NFC-Based Secure Cashless Payment Prototype | Thesis-Supporting Research Implementation

- Developed a blockchain- and NFC-based secure payment prototype with Arduino hardware experimentation.
- Implemented Arduino and Python modules for transaction processing, device-level verification, and embedded system testing.
- **Repository:** github.com/shovon-mandal/NFC | **Related Output:** ICKECS 2024 paper on Blockchain with NFC.

IOTA-Based Decentralized Trust and Transaction Verification System | Web-Based Research Prototype

- Developed a web-based prototype to demonstrate IOTA-supported payment interaction and decentralized transaction flow.
- Implemented front-end and backend components for payment demonstration, user interaction, and experimental validation.
- **Repository:** github.com/shovon-mandal/IOTA-Website | **Related Output:** ICCIT 2024 paper on IOTA Tangle.

ACADEMIC APPOINTMENTS AND TEACHING EXPERIENCE

Lecturer, Department of CSE | Northern University of Business and Technology Khulna, Bangladesh

Mar 2024 - Present

- Teach undergraduate CSE courses and laboratories in computer architecture, digital logic, OOP, technical writing, and optimization.
- Prepare course materials, assessments, lab tasks, and support student mentoring and research engagement.

Courses Taught: Computer Architecture; Digital Logic Design; Object-Oriented Programming with Java; Technical Writing; Linear Programming and Combinatorial Optimization; Digital Logic Design Lab; Linear Programming and Combinatorial Optimization Lab

Lecturer (Adjunct), Department of CSE | Imperial College of Engineering, Khulna, Bangladesh

Jul 2022 - Jun 2024

- Delivered undergraduate lectures, laboratory sessions, assessments, and academic support for core CSE courses.

Courses Taught: Database Management Systems; Computer Peripherals and Interfacing; Computer Architecture and Organization; Computer Graphics; Web Engineering; Microprocessor and Microcontrollers

Teaching Assistant, Department of CSE | *Khulna University of Engineering and Technology, Bangladesh* **Jul 2023 - Dec 2023**

- Assisted faculty members in laboratory instruction, assignment evaluation, academic record preparation, and student support.

Courses Assisted: Embedded Systems; Electronic Devices and Circuits

Student Tutor, Department of CSE | *BRAC University, Bangladesh* **Feb 2022 - Apr 2022**

- Provided individual and group academic support to undergraduate students through tutoring sessions and guided academic assistance.

Courses Assisted: Electronic Devices and Circuits

ADDITIONAL TECHNICAL PROJECTS

Student Management Android Application | *Course Project* | [GitHub Repository](#)

- Developed an Android-based student management application using Android Studio.

Embedded Hardware Prototypes | *Course and Hardware Projects*

- Built traffic-light and SIM-enabled fire-alarm prototypes using microcontroller-based interfacing.

TECHNICAL SKILLS

Programming: Python, Java, C, MATLAB, MIPS Assembly, JavaScript

Web and Database: MySQL, PHP, Laravel, HTML, CSS, JavaScript

Research or Data Tools: LaTeX, SPSS, Microsoft Excel, Google Sheets, Git, API integration, workflow automation

Systems and Hardware: Arduino prototyping, NFC payment modules, embedded systems, hardware interfacing, digital electronics

Blockchain and Security: Blockchain security, applied cryptography, secure distributed systems, IOTA-based trust mechanisms

Languages: English, Bengali (Native)

HONORS, SCHOLARSHIPS, AND ACADEMIC SERVICE

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| • Highest Distinction, BRAC University. | • Judge, Project Showcasing - CSE Fest 2025 |
| • Vice Chancellor's List, BRAC University - 5 consecutive semesters. | • Executive Member, BRAC University Computer Club. |
| • Dean's List, BRAC University - 2 consecutive semesters. | • Executive Member, Robotics Club of BRAC University. |
| • Merit-based Scholarship, BRAC University. | • Executive Member, BRACU Mongol Tori, Next Generation Mars Rover Team 4.0. |

References and supporting documents are available upon request.
